

CLASS 11-CBSE MATHEMATICS
CHAPTER: LINEAR INEQUALITIES-SET-1

TIME : 1Hr

Mx.Marks:30

SECTION A

Q.I Answer the following questions [1X3 = 3 MARKS]

1. Solve the given inequality for real x : $3(2 - x) \geq 2(1 - x)$.
2. Solve the inequality $2 \leq 3x - 4 \leq 5$
3. Solve $-12x > 30$, when
(i) x is a natural number (ii) x is an integer.

SECTION B

Q.II Answer the following questions [2X5 = 10 MARKS]

4. Solve $3x - 7 < 5 + x$, $11 - 5x \leq 1$
5. Find all pairs of consecutive even positive integers, both of which are larger than 5 such that their sum is less than 23.
6. Solve the given inequality graphically in two-dimensional plane: $x - y \leq 2$
7. Solve $\left|x + \frac{1}{3}\right| > \frac{8}{3}$
8. Solve the following inequalities and represent the solution graphically on number line:
 $3x - 7 > 2(x - 6)$, $6 - x > 11 - 2x$

SECTION C

Q.III Answer the following questions [3X3 = 9 MARKS]

9. Solve the following system of inequalities graphically: $3x + 2y \leq 12$, $x \geq 1$, $y \geq 2$
10. How many liters of water will have to be added to 1125 liters of the 45% solution of acid so that the resulting mixture will contain more than 25% but less than 30% acid content?

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11. The marks scored by Rohit in two tests were 65 and 70. Find the minimum marks he should score in the third test to have an average of at least 65 marks.

SECTION D

Q.IV Answer the following questions [4X2 = 8 MARKS]

12. Solve: $\frac{2x-3}{4} - 2 \geq \frac{4x}{3} - 6$, $2(2x + 3) < 6(x - 2) + 10$

13. IQ of a person is given by the formula

$$IQ = \frac{MA}{CA} * 100$$

Where MA is mental age and CA is chronological age. If $80 \leq IQ \leq 140$ for a group of 12 years old children, find the range of their mental age.